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Multiple victim/aggressor collision avoidance

Abstract

Certain aspects of the present disclosure provide techniques for wireless communication by a first network entity, comprising obtaining a first configuration for transmission of measurement reference signals (RS) by the first network entity, wherein the first configuration is adjusted based on information regarding one or more second configurations for transmission of measurement RS by one or more second network entities, and participating in a cross link interference (CLI) measurement procedure, in accordance with the first configuration.

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Background/Summary

BACKGROUND

Field of the Disclosure

(1) Aspects of the present disclosure relate to wireless communications, and more particularly, to techniques for avoiding collisions resulting from interference between multiple network entities.

Description of Related Art

(2) Wireless communications systems are widely deployed to provide various telecommunications services such as telephony, video, data, messaging, broadcasts, or other similar types of services. These wireless communications systems may employ multiple-access technologies capable of supporting communications with multiple users by sharing available wireless communications system resources with those users

(3) Although wireless communications systems have made great technological advancements over many years, challenges still exist. For example, complex and dynamic environments can still attenuate or block signals between wireless transmitters and wireless receivers. Accordingly, there is a continuous desire to improve the technical performance of wireless communications systems, including, for example: improving speed and data carrying capacity of communications, improving efficiency of the use of shared communications mediums, reducing power used by transmitters and receivers while performing communications, improving reliability of wireless communications, avoiding redundant transmissions and/or receptions and related processing, improving the coverage area of wireless communications, increasing the number and types of devices that can access wireless communications systems, increasing the ability for different types of devices to intercommunicate, increasing the number and type of wireless communications mediums available for use, and the like. Consequently, there exists a need for further improvements in wireless communications systems to overcome the aforementioned technical challenges and others.

SUMMARY

(4) One aspect provides a method for wireless communications by a first network entity, including obtaining a first configuration for transmission of measurement reference signals (RS) by the first network entity, wherein the first configuration is adjusted based on information regarding one or more second configurations for transmission of measurement RS by one or more second network entities; and participating in a cross link interference (CLI) measurement procedure, in accordance with the first configuration.

(5) One aspect provides a method for wireless communications by a second network entity, including obtaining information regarding a first configuration for transmission of measurement RS from a first network entity; and transmitting a request, to the first network entity, for adjustment of the first configuration, based on a comparison of the first configuration with a second configuration for transmission of measurement RS by a second network entity or another network entity.

(6) One aspect provides a method for wireless communications by a central unit (CU), including determining a first configuration for transmission of measurement RS from a first network entity, based on information regarding one or more second configurations for transmission of measurement RS from one or more second network entities; and transmitting information regarding the first configuration to the first network entity.

(7) Other aspects provide: an apparatus operable, configured, or otherwise adapted to perform the aforementioned methods as well as those described elsewhere herein; a non-transitory, computer-readable media comprising instructions that, when executed by a processor of an apparatus, cause the apparatus to perform the aforementioned methods as well as those described elsewhere herein; a computer program product embodied on a computer-readable storage medium comprising code for performing the aforementioned methods as well as those described elsewhere herein; and an apparatus comprising means for performing the aforementioned methods as well as those described elsewhere herein. By way of example, an apparatus may comprise a processing system, a device with a processing system, or processing systems cooperating over one or more networks.

(8) The following description and the appended figures set forth certain features for purposes of illustration.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The appended figures depict certain features of the various aspects described herein and are not to be considered limiting of the scope of this disclosure.

(2) FIG. **1** depicts an example wireless communications network.

(3) FIG. **2** depicts an example disaggregated base station architecture.

(4) FIG. **3** depicts aspects of an example base station and an example user equipment.

(5) FIGS. **4**A, **4**B, **4**C, and **4**D depict various example aspects of data structures for a wireless communications network.

(6) FIGS. **5**A, **5**B, and **5**C illustrates different full-duplex use cases within a wireless communication network.

(7) FIG. **6**A, FIG. **6**B, FIG. **6**C, and FIG. **6**D depict different interference scenarios occurring during full duplex (FD) communications.

(8) FIG. **7** depicts example an example of interference between an aggressor network entity and a victim network entity.

(9) FIG. **8** depicts example signaling for reference signal (RS) collision avoidance between a receiving gNB and a transmitting gNB.

(10) FIG. **9** depicts an example a call flow diagram for communications in a network between a transmitting network entity and one or more receiving network entities.

(11) FIG. **10** depicts a call flow diagram for communications in a network between a transmitting network entity and one or more receiving network entities.

(12) FIG. **11** depicts a method for wireless communications.

(13) FIG. **12** depicts a method for wireless communications.

(14) FIG. **13** depicts a method for wireless communications.

(15) FIG. **14** depicts aspects of an example communications device.

(16) FIG. **15** depicts aspects of an example communications device.

(17) FIG. **16** depicts aspects of an example communications device.

DETAILED DESCRIPTION

(18) Aspects of the present disclosure provide apparatuses, methods, processing systems, and computer-readable mediums for avoiding collisions resulting from interference between multiple network entities.

(19) The techniques may help mitigate wireless interference, for example, when a receiving network entity and a transmitting network entity communicate in a full duplex (FD) mode of communication. During FD communications, uplink (UL) and downlink (DL) transmissions may be performed simultaneously.

(20) FD capability can be present at a network entity (e.g., a base station such as a gNB), at a UE, or at both. For instance, at a UE, uplink transmissions can use one antenna panel and downlink reception may use another antenna panel. Similarly, at a gNB, uplink reception can use one panel and downlink transmission can use another panel. FD capability is typically conditional on beam separation, for example, in order to mitigate self-interference between downlink and uplink signals as well as clutter echo (e.g., where uplink transmission echoes affect uplink transmission and/or downlink reception). Benefits of FD communication include latency reduction, as it is possible to receive downlink signals in (what would be) uplink only slots. FD communication may also help improve spectrum efficiency enhancement (per cell and/or per UE), more efficient resource utilization, and coverage extension.

(21) One potential challenge with FD communication is how to manage interference caused by remote network entities, sometimes referred to as remote interference measurement (RIM). In current systems, a RIM framework may be used to help mitigate inter-gNB interference due to atmosphere ducting effects. Unfortunately, such RIM frameworks may not optimize the selected gNB beam to mitigate inter-gNB interference in case of gNB FD, and such procedures may be

limited, for example, deactivated in absence of ducting effects.

(22) For enhanced inter-gNB interference measurements, measurement reference signals (RS) can be sent from an aggressor gNB to a victim gNB in order to measure actual inter-gNB interference. In such cases, there could be multiple measurement RSs associated with multiple Tx(aggressor gNB)/Rx(victim gNB) beam pairs between aggressor gNB and victim gNB. The use of multiple measurement RSs creates a challenge, particular when there are multiple measurement RS transmissions (e.g., for n×n mutual measurements) between multiple Tx gNBs to multiple Rx gNBs.

(23) Accordingly, aspects of the present disclosure provide techniques for interference mitigation by coordinating FD communications among multiple network entities by adjusting a configuration of measurement RSs. In current wireless systems, interference mitigation for interference occurring between network entities may be limited to interference that occurs only in certain environmental conditions (e.g., where ducting effects are presents). The various techniques presented herein enhance interference management between network entities independent of certain environmental conditions to optimize FD communication.

(24) Aspects of the present disclosure provide techniques for negotiating and coordinating configurations of measurement reference signals RS between multiple network entities for interference measurement. Such techniques may allow for fewer collisions between a network entity's own transmission of measurement RSs and a neighboring network entity measurement RSs. Implementation may also allow transmission of measurement RSs to be received and efficiently measured by multiple neighboring network entities on the same RS resources to signaling reduce overhead.

Introduction to Wireless Communications Networks

(25) The techniques and methods described herein may be used for various wireless communications networks. While aspects may be described herein using terminology commonly associated with 3G, 4G, and/or 5G wireless technologies, aspects of the present disclosure may likewise be applicable to other communications systems and standards not explicitly mentioned herein.

(26) FIG. **1** depicts an example of a wireless communications network **100**, in which aspects described herein may be implemented.

(27) Generally, wireless communications network **100** includes various network entities (alternatively, network elements or network nodes). A network entity is generally a communications device and/or a communications function performed by a communications device (e.g., a user equipment (UE), a base station (BS), a component of a BS, a server, etc.). For example, various functions of a network as well as various devices associated with and interacting with a network may be considered network entities. Further, wireless communications network **100** includes terrestrial aspects, such as ground-based network entities (e.g., BSs **102**), and non-terrestrial aspects, such as satellite **140** and aircraft **145**, which may include network entities on-board (e.g., one or more BSs) capable of communicating with other network elements (e.g., terrestrial BSs) and user equipments.

(28) In the depicted example, wireless communications network **100** includes BSs **102**, UEs **104**, and one or more core networks, such as an Evolved Packet Core (EPC) **160** and 5G Core (5GC) network **190**, which interoperate to provide communications services over various communications links, including wired and wireless links.

(29) FIG. **1** depicts various example UEs **104**, which may more generally include: a cellular phone, smart phone, session initiation protocol (SIP) phone, laptop, personal digital assistant (PDA), satellite radio, global positioning system, multimedia device, video device, digital audio player, camera, game console, tablet, smart device, wearable device, vehicle, electric meter, gas pump, large or small kitchen appliance, healthcare device, implant, sensor/actuator, display, internet of things (IoT) devices, always on (AON) devices, edge processing devices, or other similar devices.

UEs **104** may also be referred to more generally as a mobile device, a wireless device, a wireless communications device, a station, a mobile station, a subscriber station, a mobile subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a remote device, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, and others.

(30) BSs **102** wirelessly communicate with (e.g., transmit signals to or receive signals from) UEs **104** via communications links **120**. The communications links **120** between BSs **102** and UEs **104** may include uplink (UL) (also referred to as reverse link) transmissions from a UE **104** to a BS **102** and/or downlink (DL) (also referred to as forward link) transmissions from a BS **102** to a UE **104**. The communications links **120** may use multiple-input and multiple-output (MIMO) antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity in various aspects.

(31) BSs **102** may generally include: a NodeB, enhanced NodeB (eNB), next generation enhanced NodeB (ng-eNB), next generation NodeB (gNB or gNodeB), access point, base transceiver station, radio base station, radio transceiver, transceiver function, transmission reception point, and/or others. Each of BSs **102** may provide communications coverage for a respective geographic coverage area **110**, which may sometimes be referred to as a cell, and which may overlap in some cases (e.g., small cell **102′** may have a coverage area **110′** that overlaps the coverage area **110** of a macro cell). A BS may, for example, provide communications coverage for a macro cell (covering relatively large geographic area), a pico cell (covering relatively smaller geographic area, such as a sports stadium), a femto cell (relatively smaller geographic area (e.g., a home)), and/or other types of cells.

(32) While BSs **102** are depicted in various aspects as unitary communications devices, BSs **102** may be implemented in various configurations. For example, one or more components of a base station may be disaggregated, including a central unit (CU), one or more distributed units (DUs), one or more radio units (RUs), a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC), or a Non-Real Time (Non-RT) RIC, to name a few examples. In another example, various aspects of a base station may be virtualized. More generally, a base station (e.g., BS **102**) may include components that are located at a single physical location or components located at various physical locations. In examples in which a base station includes components that are located at various physical locations, the various components may each perform functions such that, collectively, the various components achieve functionality that is similar to a base station that is located at a single physical location. In some aspects, a base station including components that are located at various physical locations may be referred to as a disaggregated radio access network architecture, such as an Open RAN (O-RAN) or Virtualized RAN (VRAN) architecture. FIG. **2** depicts and describes an example disaggregated base station architecture.

(33) Different BSs **102** within wireless communications network **100** may also be configured to support different radio access technologies, such as 3G, 4G, and/or 5G. For example, BSs **102** configured for 4G LTE (collectively referred to as Evolved Universal Mobile Telecommunications System (UMTS) Terrestrial Radio Access Network (E-UTRAN)) may interface with the EPC **160** through first backhaul links **132** (e.g., an S1 interface). BSs **102** configured for 5G (e.g., 5G NR or Next Generation RAN (NG-RAN)) may interface with 5GC **190** through second backhaul links **184**. BSs **102** may communicate directly or indirectly (e.g., through the EPC **160** or 5GC **190**) with each other over third backhaul links **134** (e.g., X2 interface), which may be wired or wireless.

(34) Wireless communications network **100** may subdivide the electromagnetic spectrum into various classes, bands, channels, or other features. In some aspects, the subdivision is provided based on wavelength and frequency, where frequency may also be referred to as a carrier, a subcarrier, a frequency channel, a tone, or a subband. For example, 3GPP currently defines Frequency Range 1 (FR1) as including 600 MHz-6 GHz, which is often referred to (interchangeably) as “Sub-6 GHz”. Similarly, 3GPP currently defines Frequency Range 2 (FR2) as including 26-41 GHz, which is sometimes referred to (interchangeably) as a “millimeter wave”

("mmW" or "mmWave"). A base station configured to communicate using mmWave/near mmWave radio frequency bands (e.g., a mmWave base station such as BS **180**) may utilize beamforming (e.g., **182**) with a UE (e.g., **104**) to improve path loss and range.

(35) The communications links **120** between BSs **102** and, for example, UEs **104**, may be through one or more carriers, which may have different bandwidths (e.g., 5, 10, 15, 20, 100, 400, and/or other MHz), and which may be aggregated in various aspects. Carriers may or may not be adjacent to each other. Allocation of carriers may be asymmetric with respect to DL and UL (e.g., more or fewer carriers may be allocated for DL than for UL).

(36) Communications using higher frequency bands may have higher path loss and a shorter range compared to lower frequency communications. Accordingly, certain base stations (e.g., **180** in FIG. **1**) may utilize beamforming **182** with a UE **104** to improve path loss and range. For example, BS **180** and the UE **104** may each include a plurality of antennas, such as antenna elements, antenna panels, and/or antenna arrays to facilitate the beamforming. In some cases, BS **180** may transmit a beamformed signal to UE **104** in one or more transmit directions **182′**. UE **104** may receive the beamformed signal from the base station **180** in one or more receive directions **182″**. UE **104** may also transmit a beamformed signal to the base station **180** in one or more transmit directions **182″**. BS **180** may also receive the beamformed signal from UE **104** in one or more receive directions **182′**. Base station **180** and UE **104** may then perform beam training to determine the best receive and transmit directions for each of BS **180** and UE **104**. Notably, the transmit and receive directions for BS **180** may or may not be the same. Similarly, the transmit and receive directions for UE **104** may or may not be the same.

(37) Wireless communications network **100** further includes a Wi-Fi AP **150** in communication with Wi-Fi stations (STAs) **152** via communications links **154** in, for example, a 2.4 GHz and/or 5 GHz unlicensed frequency spectrum.

(38) Certain UEs **104** may communicate with each other using device-to-device (D2D) communications link **158**. D2D communications link **158** may use one or more sidelink channels, such as a physical sidelink broadcast channel (PSBCH), a physical sidelink discovery channel (PSDCH), a physical sidelink shared channel (PSSCH), a physical sidelink control channel (PSCCH), and/or a physical sidelink feedback channel (PSFCH).

(39) EPC **160** may include various functional components, including: a Mobility Management Entity (MME) **162**, other MMEs **164**, a Serving Gateway **166**, a Multimedia Broadcast Multicast Service (MBMS) Gateway **168**, a Broadcast Multicast Service Center (BM-SC) **170**, and/or a Packet Data Network (PDN) Gateway **172**, such as in the depicted example. MME **162** may be in communication with a Home Subscriber Server (HSS) **174**. MME **162** is the control node that processes the signaling between the UEs **104** and the EPC **160**. Generally, MME **162** provides bearer and connection management.

(40) Generally, user Internet protocol (IP) packets are transferred through Serving Gateway **166**, which itself is connected to PDN Gateway **172**. PDN Gateway **172** provides UE IP address allocation as well as other functions. PDN Gateway **172** and the BM-SC **170** are connected to IP Services **176**, which may include, for example, the Internet, an intranet, an IP Multimedia Subsystem (IMS), a Packet Switched (PS) streaming service, and/or other IP services.

(41) BM-SC **170** may provide functions for MBMS user service provisioning and delivery. BM-SC **170** may serve as an entry point for content provider MBMS transmission, may be used to authorize and initiate MBMS Bearer Services within a public land mobile network (PLMN), and/or may be used to schedule MBMS transmissions. MBMS Gateway **168** may be used to distribute MBMS traffic to the BSs **102** belonging to a Multicast Broadcast Single Frequency Network (MBSFN) area broadcasting a particular service, and/or may be responsible for session management (start/stop) and for collecting eMBMS related charging information.

(42) 5GC **190** may include various functional components, including: an Access and Mobility Management Function (AMF) **192**, other AMFs **193**, a Session Management Function (SMF) **194**,

and a User Plane Function (UPF) **195**. AMF **192** may be in communication with Unified Data Management (UDM) **196**.
(43) AMF **192** is a control node that processes signaling between UEs **104** and 5GC **190**. AMF **192** provides, for example, quality of service (QoS) flow and session management.
(44) Internet protocol (IP) packets are transferred through UPF **195**, which is connected to the IP Services **197**, and which provides UE IP address allocation as well as other functions for 5GC **190**. IP Services **197** may include, for example, the Internet, an intranet, an IMS, a PS streaming service, and/or other IP services.
(45) In various aspects, a network entity or network node can be implemented as an aggregated base station, as a disaggregated base station, a component of a base station, an integrated access and backhaul (IAB) node, a relay node, a sidelink node, to name a few examples.
(46) FIG. **2** depicts an example disaggregated base station **200** architecture. The disaggregated base station **200** architecture may include one or more central units (CUs) **210** that can communicate directly with a core network **220** via a backhaul link, or indirectly with the core network **220** through one or more disaggregated base station units (such as a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC) **225** via an E2 link, or a Non-Real Time (Non-RT) RIC **215** associated with a Service Management and Orchestration (SMO) Framework **205**, or both). A CU **210** may communicate with one or more distributed units (DUs) **230** via respective midhaul links, such as an F1 interface. The DUs **230** may communicate with one or more radio units (RUs) **240** via respective fronthaul links. The RUs **240** may communicate with respective UEs **104** via one or more radio frequency (RF) access links. In some implementations, the UE **104** may be simultaneously served by multiple RUs **240**.
(47) Each of the units, e.g., the CUs **210**, the DUs **230**, the RUs **240**, as well as the Near-RT RICs **225**, the Non-RT RICs **215** and the SMO Framework **205**, may include one or more interfaces or be coupled to one or more interfaces configured to receive or transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to the communications interfaces of the units, can be configured to communicate with one or more of the other units via the transmission medium. For example, the units can include a wired interface configured to receive or transmit signals over a wired transmission medium to one or more of the other units. Additionally or alternatively, the units can include a wireless interface, which may include a receiver, a transmitter or transceiver (such as a radio frequency (RF) transceiver), configured to receive or transmit signals, or both, over a wireless transmission medium to one or more of the other units.
(48) In some aspects, the CU **210** may host one or more higher layer control functions. Such control functions can include radio resource control (RRC), packet data convergence protocol (PDCP), service data adaptation protocol (SDAP), or the like. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU **210**. The CU **210** may be configured to handle user plane functionality (e.g., Central Unit—User Plane (CU-UP)), control plane functionality (e.g., Central Unit—Control Plane (CU-CP)), or a combination thereof. In some implementations, the CU **210** can be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit can communicate bidirectionally with the CU-CP unit via an interface, such as the E1 interface when implemented in an O-RAN configuration. The CU **210** can be implemented to communicate with the DU **230**, as necessary, for network control and signaling.
(49) The DU **230** may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs **240**. In some aspects, the DU **230** may host one or more of a radio link control (RLC) layer, a medium access control (MAC) layer, and one or more high physical (PHY) layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling, modulation and demodulation, or the like) depending, at least in part, on a functional split, such as those defined by the 3.sup.rd Generation Partnership Project (3GPP). In

some aspects, the DU **230** may further host one or more low PHY layers. Each layer (or module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU **230**, or with the control functions hosted by the CU **210**.

(50) Lower-layer functionality can be implemented by one or more RUs **240**. In some deployments, an RU **240**, controlled by a DU **230**, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU(s) **240** can be implemented to handle over the air (OTA) communications with one or more UEs **104**. In some implementations, real-time and non-real-time aspects of control and user plane communications with the RU(s) **240** can be controlled by the corresponding DU **230**. In some scenarios, this configuration can enable the DU(s) **230** and the CU **210** to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

(51) The SMO Framework **205** may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework **205** may be configured to support the deployment of dedicated physical resources for RAN coverage requirements which may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework **205** may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) **290**) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs **210**, DUs **230**, RUs **240** and Near-RT RICs **225**. In some implementations, the SMO Framework **205** can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) **211**, via an O1 interface. Additionally, in some implementations, the SMO Framework **205** can communicate directly with one or more RUs **240** via an O1 interface. The SMO Framework **205** also may include a Non-RT RIC **215** configured to support functionality of the SMO Framework **205**.

(52) The Non-RT RIC **215** may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, Artificial Intelligence/Machine Learning (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC **225**. The Non-RT RIC **215** may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC **225**. The Near-RT RIC **225** may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs **210**, one or more DUs **230**, or both, as well as an O-eNB, with the Near-RT RIC **225**.

(53) In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC **225**, the Non-RT RIC **215** may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC **225** and may be received at the SMO Framework **205** or the Non-RT RIC **215** from non-network data sources or from network functions. In some examples, the Non-RT RIC **215** or the Near-RT RIC **225** may be configured to tune RAN behavior or performance. For example, the Non-RT RIC **215** may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework **205** (such as reconfiguration via **01**) or via creation of RAN management policies (such as A1 policies).

(54) FIG. **3** depicts aspects of an example BS **102** and a UE **104**.

(55) Generally, BS **102** includes various processors (e.g., **320**, **330**, **338**, and **340**), antennas **334***a-t* (collectively **334**), transceivers **332***a-t* (collectively **332**), which include modulators and demodulators, and other aspects, which enable wireless transmission of data (e.g., data source **312**) and wireless reception of data (e.g., data sink **339**). For example, BS **102** may send and receive data

between BS **102** and UE **104**. BS **102** includes controller/processor **340**, which may be configured to implement various functions described herein related to wireless communications.

(56) Generally, UE **104** includes various processors (e.g., **358**, **364**, **366**, and **380**), antennas **352***a-r* (collectively **352**), transceivers **354***a-r* (collectively **354**), which include modulators and demodulators, and other aspects, which enable wireless transmission of data (e.g., retrieved from data source **362**) and wireless reception of data (e.g., provided to data sink **360**). UE **104** includes controller/processor **380**, which may be configured to implement various functions described herein related to wireless communications.

(57) In regards to an example downlink transmission, BS **102** includes a transmit processor **320** that may receive data from a data source **312** and control information from a controller/processor **340**. The control information may be for the physical broadcast channel (PBCH), physical control format indicator channel (PCFICH), physical HARQ indicator channel (PHICH), physical downlink control channel (PDCCH), group common PDCCH (GC PDCCH), and/or others. The data may be for the physical downlink shared channel (PDSCH), in some examples.

(58) Transmit processor **320** may process (e.g., encode and symbol map) the data and control information to obtain data symbols and control symbols, respectively. Transmit processor **320** may also generate reference symbols, such as for the primary synchronization signal (PSS), secondary synchronization signal (SSS), PBCH demodulation reference signal (DMRS), and channel state information reference signal (CSI-RS).

(59) Transmit (TX) multiple-input multiple-output (MIMO) processor **330** may perform spatial processing (e.g., precoding) on the data symbols, the control symbols, and/or the reference symbols, if applicable, and may provide output symbol streams to the modulators (MODs) in transceivers **332***a*-**332***t*. Each modulator in transceivers **332***a*-**332***t* may process a respective output symbol stream to obtain an output sample stream. Each modulator may further process (e.g., convert to analog, amplify, filter, and upconvert) the output sample stream to obtain a downlink signal. Downlink signals from the modulators in transceivers **332***a*-**332***t* may be transmitted via the antennas **334***a*-**334***t*, respectively.

(60) In order to receive the downlink transmission, UE **104** includes antennas **352***a*-**352***r* that may receive the downlink signals from the BS **102** and may provide received signals to the demodulators (DEMODs) in transceivers **354***a*-**354***r*, respectively. Each demodulator in transceivers **354***a*-**354***r* may condition (e.g., filter, amplify, downconvert, and digitize) a respective received signal to obtain input samples. Each demodulator may further process the input samples to obtain received symbols.

(61) MIMO detector **356** may obtain received symbols from all the demodulators in transceivers **354***a*-**354***r*, perform MIMO detection on the received symbols if applicable, and provide detected symbols. Receive processor **358** may process (e.g., demodulate, deinterleave, and decode) the detected symbols, provide decoded data for the UE **104** to a data sink **360**, and provide decoded control information to a controller/processor **380**.

(62) In regards to an example uplink transmission, UE **104** further includes a transmit processor **364** that may receive and process data (e.g., for the PUSCH) from a data source **362** and control information (e.g., for the physical uplink control channel (PUCCH)) from the controller/processor **380**. Transmit processor **364** may also generate reference symbols for a reference signal (e.g., for the sounding reference signal (SRS)). The symbols from the transmit processor **364** may be precoded by a TX MIMO processor **366** if applicable, further processed by the modulators in transceivers **354***a*-**354***r* (e.g., for SC-FDM), and transmitted to BS **102**.

(63) At BS **102**, the uplink signals from UE **104** may be received by antennas **334***a-t*, processed by the demodulators in transceivers **332***a*-**332***t*, detected by a MIMO detector **336** if applicable, and further processed by a receive processor **338** to obtain decoded data and control information sent by UE **104**. Receive processor **338** may provide the decoded data to a data sink **339** and the decoded control information to the controller/processor **340**.

(64) Memories **342** and **382** may store data and program codes for BS **102** and UE **104**, respectively.
(65) Scheduler **344** may schedule UEs for data transmission on the downlink and/or uplink.
(66) In various aspects, BS **102** may be described as transmitting and receiving various types of data associated with the methods described herein. In these contexts, "transmitting" may refer to various mechanisms of outputting data, such as outputting data from data source **312**, scheduler **344**, memory **342**, transmit processor **320**, controller/processor **340**, TX MIMO processor **330**, transceivers **332***a-t*, antenna **334***a-t*, and/or other aspects described herein. Similarly, "receiving" may refer to various mechanisms of obtaining data, such as obtaining data from antennas **334***a-t*, transceivers **332***a-t*, RX MIMO detector **336**, controller/processor **340**, receive processor **338**, scheduler **344**, memory **342**, and/or other aspects described herein.
(67) In various aspects, UE **104** may likewise be described as transmitting and receiving various types of data associated with the methods described herein. In these contexts, "transmitting" may refer to various mechanisms of outputting data, such as outputting data from data source **362**, memory **382**, transmit processor **364**, controller/processor **380**, TX MIMO processor **366**, transceivers **354***a-t*, antenna **352***a-t*, and/or other aspects described herein. Similarly, "receiving" may refer to various mechanisms of obtaining data, such as obtaining data from antennas **352***a-t*, transceivers **354***a-t*, RX MIMO detector **356**, controller/processor **380**, receive processor **358**, memory **382**, and/or other aspects described herein.
(68) In some aspects, a processor may be configured to perform various operations, such as those associated with the methods described herein, and transmit (output) to or receive (obtain) data from another interface that is configured to transmit or receive, respectively, the data.
(69) FIGS. **4**A, **4**B, **4**C, and **4**D depict aspects of data structures for a wireless communications network, such as wireless communications network **100** of FIG. **1**.
(70) In particular, FIG. **4**A is a diagram **400** illustrating an example of a first subframe within a 5G (e.g., 5G NR) frame structure, FIG. **4**B is a diagram **430** illustrating an example of DL channels within a 5G subframe, FIG. **4**C is a diagram **450** illustrating an example of a second subframe within a 5G frame structure, and FIG. **4**D is a diagram **480** illustrating an example of UL channels within a 5G subframe.
(71) Wireless communications systems may utilize orthogonal frequency division multiplexing (OFDM) with a cyclic prefix (CP) on the uplink and downlink. Such systems may also support half-duplex operation using time division duplexing (TDD). OFDM and single-carrier frequency division multiplexing (SC-FDM) partition the system bandwidth (e.g., as depicted in FIGS. **4**B and **4**D) into multiple orthogonal subcarriers. Each subcarrier may be modulated with data. Modulation symbols may be sent in the frequency domain with OFDM and/or in the time domain with SC-FDM.
(72) A wireless communications frame structure may be frequency division duplex (FDD), in which, for a particular set of subcarriers, subframes within the set of subcarriers are dedicated for either DL or UL. Wireless communications frame structures may also be time division duplex (TDD), in which, for a particular set of subcarriers, subframes within the set of subcarriers are dedicated for both DL and UL.
(73) In FIGS. **4**A and **4**C, the wireless communications frame structure is TDD where D is DL, U is UL, and X is flexible for use between DL/UL. UEs may be configured with a slot format through a received slot format indicator (SFI) (dynamically through DL control information (DCI), or semi-statically/statically through radio resource control (RRC) signaling). In the depicted examples, a 10 ms frame is divided into 10 equally sized 1 ms subframes. Each subframe may include one or more time slots. In some examples, each slot may include 7 or 14 symbols, depending on the slot format. Subframes may also include mini-slots, which generally have fewer symbols than an entire slot. Other wireless communications technologies may have a different frame structure and/or different channels.

(74) In certain aspects, the number of slots within a subframe is based on a slot configuration and a numerology. For example, for slot configuration 0, different numerologies (μ) 0 to 5 allow for 1, 2, 4, 8, 16, and 32 slots, respectively, per subframe. For slot configuration 1, different numerologies 0 to 2 allow for 2, 4, and 8 slots, respectively, per subframe. Accordingly, for slot configuration 0 and numerology μ, there are 14 symbols/slot and 2μ slots/subframe. The subcarrier spacing and symbol length/duration are a function of the numerology. The subcarrier spacing may be equal to 2.sup.μ×15 kHz, where is the numerology 0 to 5. As such, the numerology μ=0 has a subcarrier spacing of 15 kHz and the numerology μ=5 has a subcarrier spacing of 480 kHz. The symbol length/duration is inversely related to the subcarrier spacing. FIGS. **4**A, **4**B, **4**C, and **4**D provide an example of slot configuration 0 with 14 symbols per slot and numerology μ=2 with 4 slots per subframe. The slot duration is 0.25 ms, the subcarrier spacing is 60 kHz, and the symbol duration is approximately 16.67 s.
(75) As depicted in FIGS. **4**A, **4**B, **4**C, and **4**D, a resource grid may be used to represent the frame structure. Each time slot includes a resource block (RB) (also referred to as physical RBs (PRBs)) that extends, for example, 12 consecutive subcarriers. The resource grid is divided into multiple resource elements (REs). The number of bits carried by each RE depends on the modulation scheme.
(76) As illustrated in FIG. **4**A, some of the REs carry reference (pilot) signals (RS) for a UE (e.g., UE **104** of FIGS. **1** and **3**). The RS may include demodulation RS (DMRS) and/or channel state information reference signals (CSI-RS) for channel estimation at the UE. The RS may also include beam measurement RS (BRS), beam refinement RS (BRRS), and/or phase tracking RS (PT-RS).
(77) FIG. **4**B illustrates an example of various DL channels within a subframe of a frame. The physical downlink control channel (PDCCH) carries DCI within one or more control channel elements (CCEs), each CCE including, for example, nine RE groups (REGs), each REG including, for example, four consecutive REs in an OFDM symbol.
(78) A primary synchronization signal (PSS) may be within symbol **2** of particular subframes of a frame. The PSS is used by a UE (e.g., **104** of FIGS. **1** and **3**) to determine subframe/symbol timing and a physical layer identity.
(79) A secondary synchronization signal (SSS) may be within symbol **4** of particular subframes of a frame. The SSS is used by a UE to determine a physical layer cell identity group number and radio frame timing.
(80) Based on the physical layer identity and the physical layer cell identity group number, the UE can determine a physical cell identifier (PCI). Based on the PCI, the UE can determine the locations of the aforementioned DMRS. The physical broadcast channel (PBCH), which carries a master information block (MIB), may be logically grouped with the PSS and SSS to form a synchronization signal (SS)/PBCH block. The MIB provides a number of RBs in the system bandwidth and a system frame number (SFN). The physical downlink shared channel (PDSCH) carries user data, broadcast system information not transmitted through the PBCH such as system information blocks (SIBs), and/or paging messages.
(81) As illustrated in FIG. **4**C, some of the REs carry DMRS (indicated as R for one particular configuration, but other DMRS configurations are possible) for channel estimation at the base station. The UE may transmit DMRS for the PUCCH and DMRS for the PUSCH. The PUSCH DMRS may be transmitted, for example, in the first one or two symbols of the PUSCH. The PUCCH DMRS may be transmitted in different configurations depending on whether short or long PUCCHs are transmitted and depending on the particular PUCCH format used. UE **104** may transmit sounding reference signals (SRS). The SRS may be transmitted, for example, in the last symbol of a subframe. The SRS may have a comb structure, and a UE may transmit SRS on one of the combs. The SRS may be used by a base station for channel quality estimation to enable frequency-dependent scheduling on the UL.
(82) FIG. **4**D illustrates an example of various UL channels within a subframe of a frame. The

PUCCH may be located as indicated in one configuration. The PUCCH carries uplink control information (UCI), such as scheduling requests, a channel quality indicator (CQI), a precoding matrix indicator (PMI), a rank indicator (RI), and HARQ ACK/NACK feedback. The PUSCH carries data, and may additionally be used to carry a buffer status report (BSR), a power headroom report (PHR), and/or UCI.

Example Full Duplex Use Cases

(83) Aspects of the present disclosure relate to wireless communications, and more particularly, to techniques for adjusting, based on communication with one or more network entities a configuration for transmission of measurement reference signals (RSs) in order to mitigate inter-gNB interference with coordination.

(84) As noted above, in some cases, wireless communication devices, such as UEs and BSs, may communicate using multiple antenna panels. In some cases, the multiple antenna panels may be used for half-duplex (HD) communication, such as in current 5G new radio (NR) communication systems, in which downlink (DL) and uplink (UL) transmissions are transmitted non-simultaneously (e.g., transmitted in different time resources). HD communication may be considered baseline behavior in Release 15 (R-15) and 16 (R-16) of 5G NR. In other cases, the use of multiple antenna panels may allow for full duplex (FD) communication whereby uplink (UL) and downlink (DL) transmissions may be performed simultaneously (e.g., in the same time resources). For example, in some cases, UL transmission by the UE may be performed on one panel while DL reception may be performed simultaneously on another panel of the UE. Likewise, at a BS, DL transmission by the BS may be performed on one antenna panel while UL reception may be performed on another antenna panel.

(85) FD capability may be conditioned on beam separation (e.g., frequency separation or spatial separation) and may still be subject to certain self-interference between UL and DL (e.g., UL transmission directly interferes with DL reception) as well as clutter echo (e.g., where UL transmission echoes affect UL transmission and/or DL reception). However, while FD capability may be subject to certain interference, FD capability provides for reduced transmission and reception latency (e.g., it may be possible to receive DL transmissions in an UL-only slot), increased spectrum efficiency (e.g., per cell and/or per UE), and more efficient resource utilization.

(86) FIGS. **5**A-**5**C illustrates different FD use cases within a wireless communication network, such as the wireless communication network **100**. For example, FIG. **5**A illustrates a first FD use case involving transmission between one UE **502** and two base stations (or multiple transmission reception points (mTRP)), BS **504** and BS **506**. In some cases, UE **502** may be representative of UE **104** of FIG. **1** and BSs **504**, **506** may be representative of BS **102** of FIG. **1**. As shown, the UE **502** may simultaneously receive DL transmissions **508** from the BS **504** and transmit UL transmissions **510** to the BS **506**. In some cases, the DL transmissions **508** and UL transmissions **510** may be performed using different antenna panels to facilitate the simultaneous transmission and reception.

(87) A second FD use case is illustrated in FIG. **5**B involving two different UEs and one BS. As illustrated, the UE **502** may receive DL transmissions **508** from the BS **504** while another UE **512** may simultaneously transmit UL transmission **510** to the BS **504**. Thus, in this example, BS **504** is conducting simultaneous uplink and downlink communications.

(88) A third FD use case is illustrated in FIG. **5**C involving one BS and one UE. As illustrated, the UE **502** may receive DL transmissions **508** from the BS **504** and may simultaneously transmit UL transmissions **510** to the BS **504**. As noted above, such simultaneous reception/transmission by the UE **502** may be facilitated by different antenna panels.

(89) Table 1, below, illustrates various example scenarios in which each of the FD use cases may be used.

(90) TABLE-US-00001 TABLE 1 Base Station UE FD use case FD disabled FD disabled Baseline R-15/16 5G behavior FD disabled FD enabled Use case #1 (FIG. 5A) for mTRP FD enabled FD

disabled Use case #2 (FIG. 5B) + R-16 IAB FD enabled FD enabled Use case #3 (FIG. 5C)

(91) As shown in Table 1, if FD capability is disabled at both the base station and UE, the baseline R-15 and R-16 5G behavior may be used (e.g., HD communication). If FD capability is disabled at the BS but enabled at the UE, the UE may operate according to the first example FD use case shown in FIG. **5**A in which the UE may communicate with two different TRPs simultaneously (e.g., simultaneous UL and DL transmissions) using two different antenna panels. If FD is enabled at the BS but disabled at the UE (e.g., the UE is not capable of FD), the BS may operate according to the second example FD use case shown in FIG. **5**B in which the BS may communicate with two different UEs simultaneously (e.g., simultaneous UL and DL transmissions) using two different antenna panels. Finally, if FD is enabled at both the BS and the UE, the BS and UE may operate according to the third example FD use case shown in FIG. **5**C in which the BS and UE may communicate with each other simultaneously on the UL and DL, each of the BS and UE using different antenna panels for UL and DL transmissions.

(92) FIG. **6**A, FIG. **6**B, FIG. **6**C, and FIG. **6**D illustrate example CLI scenarios for various FD communication use cases.

(93) As illustrated in FIG. **6**A, a first CLI scenario may occur when FD is enabled for a gNB but disabled for each connected UE (which in turn may be enabled for half-duplex (HD) communication). The gNB may communicate using FD capabilities according to Use Case **2** of FIG. **5**B. In this case, interference may include CLI between UEs, SI from the FD gNB, and CLI between the gNB and neighboring gNBs.

(94) As illustrated in FIG. **6**B, a second CLI scenario may occur when FD is enabled for both a gNB and an FD UE/customer premise equipment (CPE) connected to the gNB. The gNB may communicate with the FD UE using FD capabilities according to Use Case **3** of FIG. **5**C. If the gNB is connected to a HD UE alongside the FD UE, the gNB communicates with the HD UE according to Use Case **2** of FIG. **5**B. In this case, interference may include CLI between UEs, SI from the gNB and the FD UE, and CLI between the FD gNB and neighboring gNBs.

(95) As illustrated in FIG. **6**C, a third CLI scenario may occur when FD is disabled for two gNBs (e.g., in a multiple TRP scenario) and enabled at one UE/CPE connected to the two gNBs. The two gNBs may communicate with the FD UE using FD capabilities according to Use Case **1** of FIG. **5**A. If one of the two gNBs is connected to a HD UE alongside the FD UE, the one gNB communicates with both the HD UE and the FD UE. In this case, interference may include CLI between UEs, SI from the FD UE, and CLI between the two gNBs.

(96) As illustrated in FIG. **6**D, a fourth CLI scenario may occur when two FD IAB nodes have conditional enhanced duplexing capability. In the illustrated example, each of the two IAB nodes are connected to a parent node. Each IAB node may be enabled for FD communication and may communicate with at least two UEs according to Use Case **2** of FIG. **5**B. In some cases, nodes involved in the FD use case depicted in FIG. **6**D may support same frequency full duplex (SFFD) and frequency division multiplexing (FDM)/spatial division multiplexing (SDM) with resource block group (RBG) granularity. In this case, interference may include CLI between IAB nodes and SI from each IAB node.

(97) As noted above, FD communication may be facilitated through the use of FDM or SDM. In FDM, the simultaneous UL and DL transmissions may be transmitted in the same time resources, but on separate frequency bands separated by some guard band. In SDM, the simultaneous UL and DL transmissions may transmitted on the same time and frequency resources but spatially separated into different, directional transmission beams. Such FD communication contrasts with HD communication that uses time division multiplexing (TDM), in order to schedule UL and DL transmissions on the same or different frequency resources, but at different times.

Aspects Related to Collision Avoidance for Multiple Victims and/or Aggressors

(98) Aspects of the present disclosure provide techniques for coordinating FD communications among multiple network entities by adjusting a configuration of measurement reference signals

(RSs). The various techniques presented herein may allow for inter-gNB interference management and could help optimize FD and TDD communication sessions.
(99) As noted above, remote interference measurement (RIM) frameworks may be deployed to help mitigate inter-gNB interference that occurs as a result of atmosphere ducting effects. Atmospheric ducting occurs in lower layers of Earth's atmosphere, where waves are bent by atmospheric refraction resulting from particular weather conditions (e.g., inversion). Atmospheric refraction causes electromagnetic waves like those deployed in a wireless system to travel over far greater distances than the distances configured for deployment under normal weather conditions. As a result, signaling from a network entity subject to ducting effect may travel father than anticipated, and may interfere with other network entities along its extended path.
(100) FIG. **7** illustrates inter-gNB interference that may occur in various FD deployment scenarios, including deployment scenarios affected and unaffected by ducting. An aggressor gNB may have downlink transmissions “D” scheduled in coordination with downlink transmission of a victim gNB, as illustrated in the first and last rows of FIG. **7**. Coordination may be enabled using a transmission delay at the aggressor gNB, allowing alignment of aggressor and victim transmissions to mitigate interference. However, propagation delays (e.g., 3 symbol, 5 symbol, or 14 symbol delays) can cause the aggressor transmission to arrive later in time, as illustrated in rows **2** through **4** of FIG. **7**. As a result, downlink transmissions by the aggressor gNB cause interference with uplink transmissions “U” of the victim gNB. Interference may cause signaling failure and wasted resources.
(101) Conventionally, RIM frameworks only optimize a selected gNB beam to mitigate inter-gNB interference during FD communication when a ducting effect is present. Such RIM frameworks typically provide a single RS to mitigate beam interference due to the ducting effect, but do not provide per-beam RSs. As noted above, the procedure may limited for RIM in some cases (e.g., deactivated in absence of ducting effect).
(102) Aspects of the present disclosure provide enhanced inter-gNB interference measurements that may help enable per-beam optimization to mitigate inter-gNB interference during FD communication in an expanded range of deployment scenarios (e.g., non-ducting scenarios). For example, the techniques described herein may help coordinate the configuration of measurement RSs sent from an aggressor network entity (e.g., a transmitting gNB) to a victim network entity (e.g., a receiving gNB) to measure actual inter-gNB interference. In some aspects, there may be multiple measurement RSs associated with multiple transmitting/receiving gNB beam pairs between the aggressor network entity and the victim network entity.
(103) The techniques described herein may help coordinate configuration of measurement RS for “n×n” mutual measurements between multiple transmitting network entities to multiple receiving network entity for interference measurement. Such coordination may allow avoid collisions, for example of reception of measurement RS transmissions from gNBs neighboring one gNB with that gNB's own measurement RS transmissions. The coordination may also allow measurement RSs to be received and measured by multiple neighbor gNBs on the same resource to reduce signaling overhead. Implementation may also allow a gNB to receive and monitor (listen to) a set of measurement RSs from multiple neighbor aggressor gNBs operating in a time division multiplexed (TDM), code-division multiplexed (CDM), and/or frequency division multiplexed (FDM) mode.
(104) The call flow diagram **800** of FIG. **8** illustrates how RS measurement configurations for multiple transmitting (Tx) and receiving (Rx) gNBs may be coordinated.
(105) In the illustrated example, a Tx gNB begins by sending an indication of its configuration of measurement RS to a Rx gNB. Based on this information (regarding the RS measurement configuration for the Tx gNB), the Rx gNB may adjust its own configuration, if possible, to avoid collision. For example, if possible, the Rx gNB may adjust when its measurement RS transmissions occur and/or what spatial resources (e.g., beams) are used for such transmission.
(106) If the Rx gNB is not able to adjust its configuration to avoid collision, the Rx gNB may send

a request for the Tx gNB to adjust its configuration. As illustrated, in such cases, the Tx gNB may send an acknowledgment (ACK/NACK) to the Rx gNB (indicating whether the request is accepted or not). Based on the acknowledgment, the Rx gNB may finalize its measurement RS configuration. The Tx gNB and Rx gNB may schedule their transmission and reception of measurement RS, based on the negotiation.

(107) FIGS. **9** and **10** illustrate call flow diagrams (**900** and **1000**) for how information regarding a measurement RS configuration may be signaled explicitly (FIG. **9**) or may be sensed (FIG. **10**).

(108) The Tx and Rx network entities may be examples of the BS **102** depicted and described with respect to FIGS. **1** and **3**. In some cases, the Tx and Rx network entities may be an example of the CU **210**, the DU **230**, and/or the RU **240** depicted and described with respect to FIG. **2**. However, the Tx and Rx network entities may be another type of network entity or network node, such as those described herein.

(109) Referring first to the call flow diagram **900** of FIG. **9**, at **902**, the Tx network entity sends information regarding a first configuration for transmission of measurement RS to one or more receiving network entities. For example, the information may include information regarding configured measurement RS periodicity, measurement RS patterns, and/or configured measurement RS resources.

(110) As noted above, in some cases, an Rx network entity (e.g., Rx gNB) may adjust its own measurement RS configuration, based on the information it receives from the Tx network entity (e.g., Tx gNB) in order to avoid collisions. In some cases, the Rx network entity may not be able to change its configuration and may need to ask the Tx network entity to modify its configuration.

(111) For example, at **904**, one of the Rx network entities may send a request for the Tx network entity to adjust its configuration. In the illustrated example, at **906**, the Tx network entity adjusts its configuration in response to the request. In some cases, the Tx network entity may send an acknowledgment (ACK) of the request, confirming it is able to adjust its configuration (and possibly indicating/confirming the adjusted configuration).

(112) At **908**, the Tx and Rx network entities perform the CLI measurement procedure, based on the adjusted configuration(s). In some cases, the CLI measurement procedure comprises scheduled measurement RS transmissions, in accordance with the first configuration and corresponding measurement RS reception and reporting. In some cases, a Tx network entity may adjust the first configuration according to action performed by a CU (e.g., the CU **210** of FIG. **2**).

(113) As noted above, if other network entities (e.g. one or more Rx gNBs) do not have a collision, or if one or more receiving network entities cannot adjust to avoid a collision, the one or more receiving network entities may signal their own info and request that the transmitting network entity adjust the measurement RS configuration.

(114) Based on the negotiation and coordination between the transmitting network entity and the one or more receiving network entities discussed above, the network entities may finalize their own configuration of measurement RS and transmit, receive, and measure based on the configuration.

(115) As noted above, transmitting and receiving network entities (e.g., gNBs) may be an example of the BS **102** depicted and described with respect to FIGS. **1** and **3**. In some cases, the transmitting and receiving network entities may be examples of the CU **210**, the DU **230**, and/or the RU **240** depicted and described with respect to the disaggregated network of FIG. **2**. In such cases, a CU may obtain configuration information from its DU(s) via a backhaul interface and adjust one or more of the configurations accordingly.

(116) In one case, a CU may determine a first configuration for transmission of measurement RSs from Tx network entity, based on information regarding one or more second configurations for transmission of measurement RS from one or more Rx network entities. The CU may then transmit information regarding the first configuration to the first network entity.

(117) If the gNBs/DUs participating in the negotiation and coordination belong to the same CU, the network may require DU-CU F1 signaling for negotiation and coordination. In some cases, the CU

may make certain decisions on behalf of the transmitting and/or receiving network entities. In one example, a CU may coordinate measurement RS configurations for multiple entities in order to avoid collision and reuse measurement RSs of a Tx gNB as measured by multiple Rx gNBs. In such cases, the CU may coordinate at least one of a measurement RS periodicity, measurement RS pattern, and/or measurement RS resources to avoid collisions or allow for multiple entities to perform measurement at the same time.
(118) If the gNBs/DUs belong to different CUs, the network may utilize signaling between CUs (e.g., CU-CU Xn signaling) for negotiation and coordination. Based on coordination by the involved CUs, the CUs may make certain decisions on behalf of the transmitting and/or receiving network entities. In one example, the CUs may negotiate and determine how to avoid collision and reuse measurement RSs of a Tx gNB as measured by multiple Rx gNBs.
(119) According to certain aspects, negotiation and coordination may be performed via backhaul signaling and/or over-the-air (OTA) signaling.
(120) As shown in the call flow diagram **1000** of FIG. **10**, CLI coordination and negotiation may be based on mutual detectability and sensing of measurement RS resource of multiple network entities. In this option, there may be preconfigured multiple candidate resource locations and/or windows to transmit measurement RSs by different network entities. In certain cases, there may be preconfigured multiple candidate resource locations to transmit measurement RSs by different network entities.
(121) Before each network entity transmits its own measurement RSs, at **1002**, they may first perform sensing operations. For example, a gNB may observe whether recent candidate resource locations are occupied. This may allow a sensing gNB to select certain possibly unoccupied candidate resource locations to transmit measurement RSs. As a result, the sensing gNB may better avoid collision with other nearby neighbor gNBs, thus mitigating interference and resultant waste.
(122) After detecting/sensing candidate measurement RS resources, the network entities may proceed in a similar manner as described above with reference to FIG. **9**.
(123) For example, at **1004**, the Tx network entity may adjust its own RS configuration based on detected information regarding one or more second configurations. At **1008**, the Tx and Rx network entities perform the CLI measurement procedure, based on the (sensed/detected) adjusted measurement RS configuration(s).
(124) In some cases, one or more rules (e.g., specified in a standard) may define a sensing time duration, periodicity, and/or offset. Each network entity may follow such rules to determine when to start transmitting its measurement RS.
(125) Similarly, one or more rules (e.g., specified in a standard) may define certain periodic or aperiodic time windows for reception of measurement RS. For example, when a gNB (or other network entity) joins a network, that gNB may be allocated a certain time window to receive and measure a set of measurement RSs transmitted from other neighbor gNBs for inter-gNB interference measurement.
(126) In some cases, while sensing/detecting candidate measurement RSs, a network entity may listen to several frames while maintaining periodicity. This sensing period may be similar to a vehicle to everything (V2X) information exchange. After the sensing period ends, the network entity may update measurement RSs or repeat the sensing procedure.
(127) In some cases, a network entity may request reconfiguration of receiving measurement windows. In some cases, if environmental changes occur or a number of network entities change, a gNB may adjust periodicity. For example, if a number of gNBs are added to the network, a request may be made to reconfigure Rx measurement windows to reduce periodicity to maintain a same overhead (as before the gNBs were added).
(128) Using the techniques described herein, interference measurement RS configurations may be negotiated such that reception periods of measurement RSs of neighboring network entities will not collide with transmission periods from those same entities. In some cases, the negotiations may

attempt to avoid a network entity receive multiple colliding measurement RSs transmitted from other network entities.
(129) Aspects of the present disclosure provide techniques for negotiation and coordination between network entities to avoid collision that causes wasteful interference during FD communication. Aspects may be applied in a network as depicted in FIG. **1**, or in a disaggregated network as depicted in FIG. **2**. Implementation of techniques described herein may allow transmission of measurement RSs to be received and measured by multiple neighboring network entities at the same resource to reduce overhead. As a result, collisions may be reduced along with signal failure and excessive retransmission.
Example Operations of a Network Entity
(130) FIG. **11** shows a method **1100** for wireless communications by a first network entity, such as BS **102** of FIGS. **1** and **3**, or a disaggregated base station as discussed with respect to FIG. **2**.
(131) Method **1100** begins at **1105** with obtaining a first configuration for transmission of measurement RS by the first network entity, wherein the first configuration is adjusted based on information regarding one or more second configurations for transmission of measurement RS by one or more second network entities. In some cases, the operations of this step refer to, or may be performed by, measurement RS configuration circuitry as described with reference to FIG. **14**.
(132) Method **1100** then proceeds to step **1110** with participating in a CLI measurement procedure, in accordance with the first configuration. In some cases, the operations of this step refer to, or may be performed by, CLI measurement procedure circuitry as described with reference to FIG. **14**.
(133) Various aspects relate to the method **1100**, including the following aspects.
(134) In some aspects, participating in the CLI measurement procedure comprises at least one of: scheduling measurement RS transmissions, in accordance with the first configuration; or scheduling measurement RS reception, in accordance with the first configuration.
(135) In some aspects, the first configuration and the one or more second configurations indicate at least one of a measurement RS periodicity, measurement RS pattern, or measurement RS resources.
(136) In some aspects, method **1100** further includes adjusting the first configuration, based on the information regarding the one or more second configurations, to avoid collision of measurement RS transmissions from the first network entity with RS transmissions from the one or more second network entities.
(137) In some aspects, method **1100** further includes adjusting the first configuration, based on the information regarding the one or more second configurations, to enable measurement RS transmissions from the first network entity to be received by multiple of the one or more second network entities.
(138) In some aspects, method **1100** further includes obtaining the information regarding the one or more second configurations from the one or more second network entities. In some aspects, method **1100** further includes transmitting information regarding the first configuration to one or more of the second network entities. In some aspects, method **1100** further includes receiving a request, from one or more of the second network entities, to adjust the first configuration. In some aspects, method **1100** further includes adjusting the first configuration in response to the request.
(139) In some aspects, the first and second network entities comprise a first DU and a second DU. In some aspects, obtaining the first configuration involves signaling between the first DU and a CU via a backhaul interface, wherein the first DU belongs to the CU. In some aspects, the CU adjusts the first configuration based on the information regarding the one or more second configurations for transmission of measurement RS by one or more second network entities. In some aspects, if the first DU and the second DU belong to different CUs, obtaining the first configuration also involves signaling between the different CUs to obtain the information regarding the one or more second configurations for transmission of measurement RS by one or more second network entities.
(140) In some aspects, method **1100** further includes obtaining the information regarding the one or

more second configurations by monitoring candidate measurement RS resources. In some aspects, participating in the CLI measurement procedure comprises transmitting measurement RS on a candidate measurement RS resources after determining that candidate measurement RS resource is unoccupied by any of the one or more second network entities. In some aspects, the first network entity monitors candidate measurement resources within time windows allocated for CLI measurement. In some aspects, the time windows are defined by at least one of a sensing time duration, periodicity, or offset for transmission of measurement RS.
(141) In one aspect, method **1100**, or any aspect related to it, may be performed by an apparatus, such as communications device **1400** of FIG. **14**, which includes various components operable, configured, or adapted to perform the method **1100**. Communications device **1400** is described below in further detail.
(142) Note that FIG. **11** is just one example of a method, and other methods including fewer, additional, or alternative steps are possible consistent with this disclosure.
(143) FIG. **12** shows a method **1200** for wireless communications by a second network entity, such as BS **102** of FIGS. **1** and **3**, or a disaggregated base station as discussed with respect to FIG. **2**.
(144) Method **1200** begins at **1205** with obtaining information regarding a first configuration for transmission of measurement RS from a first network entity. In some cases, the operations of this step refer to, or may be performed by, configuration information circuitry as described with reference to FIG. **15**.
(145) Method **1200** then proceeds to step **1210** with transmitting a request, to the first network entity, for adjustment of the first configuration, based on a comparison of the first configuration with a second configuration for transmission of measurement RS by a second network entity or another network entity. In some cases, the operations of this step refer to, or may be performed by, configuration adjustment request circuitry as described with reference to FIG. **15**.
(146) Various aspects relate to the method **1200**, including the following aspects.
(147) In some aspects, the first configuration and the one or more second configurations indicate at least one of a measurement RS periodicity, measurement RS pattern, or measurement RS resources. In some aspects, the request is for adjustment of the first configuration to avoid collision of measurement RS transmissions from the first network entity with RS transmissions from the second network entity or the other network entity. In some aspects, method **1200** further includes obtaining the information regarding the first configuration from the first network entity.
(148) In some aspects, the first and second network entities comprise a first DU and a second DU. In some aspects, obtaining the first configuration involves signaling between the first DU and a CU, wherein at least the second DU belongs to the CU. In some aspects, the CU adjusts the first configuration based on the information regarding the one or more second configurations for transmission of measurement RS by one or more second network entities. In some aspects, if the first DU and second DU belong to different CUs, obtaining the first configuration also involves signaling between the different CUs to obtain the information regarding the one or more second configurations for transmission of measurement RS by one or more second network entities.
(149) In some aspects, obtaining the information regarding the first configuration comprises monitoring candidate measurement RS resources. In some aspects, the second network entity monitors candidate measurement resources within time windows allocated for CLI measurement. In some aspects, the time windows are defined by at least one of a sensing time duration, periodicity, or offset for transmission of measurement RS.
(150) In one aspect, method **1200**, or any aspect related to it, may be performed by an apparatus, such as communications device **1500** of FIG. **15**, which includes various components operable, configured, or adapted to perform the method **1200**. Communications device **1500** is described below in further detail.
(151) Note that FIG. **12** is just one example of a method, and other methods including fewer, additional, or alternative steps are possible consistent with this disclosure.

Example Operations of a Central Unit

(152) FIG. **13** shows a method **1300** for wireless communications by a CU, such as BS **102** of FIGS. **1** and **3**, or a disaggregated base station as discussed with respect to FIG. **2**.

(153) Method **1300** begins at **1305** with determining a first configuration for transmission of measurement RS from a first network entity, based on information regarding one or more second configurations for transmission of measurement RS from one or more second network entities. In some cases, the operations of this step refer to, or may be performed by, network configuration determination circuitry as described with reference to FIG. **16**.

(154) Method **1300** then proceeds to step **1310** with transmitting information regarding the first configuration to the first network entity. In some cases, the operations of this step refer to, or may be performed by, network configuration indication circuitry as described with reference to FIG. **16**.

(155) Various aspects relate to the method **1300**, including the following aspects.

(156) In some aspects, method **1300** further includes transmitting information regarding the second configuration to the second network entity. In some aspects, the first configuration and the one or more second configurations indicate at least one of a measurement RS periodicity, measurement RS pattern, or measurement RS resources. In some aspects, determining the first configuration comprises adjusting the first configuration to avoid collision of measurement RS transmissions from the first network entity with RS transmissions from the one or more second network entities. In some aspects, determining the first configuration comprises adjusting the first configuration to enable measurement RS transmissions from the first network entity to be received by multiple of the one or more second network entities. In some aspects, the first and second network entities comprise first and second DU.

(157) In one aspect, method **1300**, or any aspect related to it, may be performed by an apparatus, such as communications device **1600** of FIG. **16**, which includes various components operable, configured, or adapted to perform the method **1300**. Communications device **1600** is described below in further detail.

(158) Note that FIG. **13** is just one example of a method, and other methods including fewer, additional, or alternative steps are possible consistent with this disclosure.

Example Communications Devices

(159) FIG. **14** depicts aspects of an example communications device **1400**. In some aspects, communications device **1400** is a first network entity, such as BS **102** described above with respect to FIGS. **1** and **3**.

(160) The communications device **1400** includes a processing system **1405** coupled to the transceiver **1465** (e.g., a transmitter and/or a receiver) and/or a network interface **1475**. The transceiver **1465** is configured to transmit and receive signals for the communications device **1400** via the antenna **1470**, such as the various signals as described herein. The network interface **1475** is configured to obtain and send signals for the communications device **1400** via communication link(s), such as a backhaul link, midhaul link, and/or fronthaul link as described herein, such as with respect to FIG. **2**. The processing system **1405** may be configured to perform processing functions for the communications device **1400**, including processing signals received and/or to be transmitted by the communications device **1400**.

(161) The processing system **1405** includes one or more processors **1410**. In various aspects, one or more processors **1410** may be representative of one or more of receive processor **338**, transmit processor **320**, TX MIMO processor **330**, and/or controller/processor **340**, as described with respect to FIG. **3**. The one or more processors **1410** are coupled to a computer-readable medium/memory **1435** via a bus **1460**. In certain aspects, the computer-readable medium/memory **1435** is configured to store instructions (e.g., computer-executable code) that when executed by the one or more processors **1410**, cause the one or more processors **1410** to perform the method **1100** described with respect to FIG. **11**, or any aspect related to it. Note that reference to a processor of communications device **1400** performing a function may include one or more processors **1410** of

communications device **1400** performing that function.
(162) In the depicted example, the computer-readable medium/memory **1435** stores code (e.g., executable instructions), such as measurement RS configuration code **1440**, CLI measurement procedure code **1445**, configuration adjustment code **1450**, and network information management code **1455**. Processing of the measurement RS configuration code **1440**, CLI measurement procedure code **1445**, configuration adjustment code **1450**, and network information management code **1455** may cause the communications device **1400** to perform the method **1100** described with respect to FIG. **11**, or any aspect related to it.
(163) The one or more processors **1410** include circuitry configured to implement (e.g., execute) the code stored in the computer-readable medium/memory **1435**, including circuitry such as measurement RS configuration circuitry **1415**, CLI measurement procedure circuitry **1420**, configuration adjustment circuitry **1425**, and network information management circuitry **1430**. Processing with measurement RS configuration circuitry **1415**, CLI measurement procedure circuitry **1420**, configuration adjustment circuitry **1425**, and network information management circuitry **1430** may cause the communications device **1400** to perform the method **1100** as described with respect to FIG. **11**, or any aspect related to it.
(164) Various components of the communications device **1400** may provide means for performing the method **1100** as described with respect to FIG. **11**, or any aspect related to it. Means for transmitting, sending or outputting for transmission may include transceivers **332** and/or antenna(s) **334** of the BS **102** illustrated in FIG. **3** and/or the transceiver **1465** and the antenna **1470** of the communications device **1400** in FIG. **14**. Means for receiving or obtaining may include transceivers **332** and/or antenna(s) **334** of the BS **102** illustrated in FIG. **3** and/or the transceiver **1465** and the antenna **1470** of the communications device **1400** in FIG. **14**.
(165) According to some aspects, measurement RS configuration circuitry **1415** obtains a first configuration for transmission of measurement RS by the first network entity, wherein the first configuration is adjusted based on information regarding one or more second configurations for transmission of measurement RS by one or more second network entities.
(166) According to some aspects, CLI measurement procedure circuitry **1420** participates in a CLI measurement procedure, in accordance with the first configuration. In some aspects, participating in the CLI measurement procedure comprises at least one of: scheduling measurement RS transmissions, in accordance with the first configuration; or scheduling measurement RS reception, in accordance with the first configuration. In some aspects, the first configuration and the one or more second configurations indicate at least one of a measurement RS periodicity, measurement RS pattern, or measurement RS resources.
(167) According to some aspects, configuration adjustment circuitry **1425** adjusts the first configuration, based on the information regarding the one or more second configurations, to avoid collision of measurement RS transmissions from the first network entity with RS transmissions from the one or more second network entities. In some examples, configuration adjustment circuitry **1425** adjusts the first configuration, based on the information regarding the one or more second configurations, to enable measurement RS transmissions from the first network entity to be received by multiple of the one or more second network entities.
(168) According to some aspects, network information management circuitry **1430** obtains the information regarding the one or more second configurations from the one or more second network entities. In some examples, network information management circuitry **1430** transmits information regarding the first configuration to one or more of the second network entities.
(169) In some examples, configuration adjustment circuitry **1425** receives a request, from one or more of the second network entities, to adjust the first configuration. In some examples, configuration adjustment circuitry **1425** adjusts the first configuration in response to the request.
(170) In some aspects, the first and second network entities comprise a first DU and a second DU. In some aspects, obtaining the first configuration involves signaling between the first DU and a CU

via a backhaul interface, wherein the first DU belongs to the CU. In some aspects, the CU adjusts the first configuration based on the information regarding the one or more second configurations for transmission of measurement RS by one or more second network entities. In some aspects, if the first DU and the second DU belong to different CUs, obtaining the first configuration also involves signaling between the different CUs to obtain the information regarding the one or more second configurations for transmission of measurement RS by one or more second network entities.

(171) In some examples, network information management circuitry **1430** obtains the information regarding the one or more second configurations by monitoring candidate measurement RS resources. In some aspects, participating in the CLI measurement procedure comprises transmitting measurement RS on a candidate measurement RS resources after determining that candidate measurement RS resource is unoccupied by any of the one or more second network entities. In some aspects, the first network entity monitors candidate measurement resources within time windows allocated for CLI measurement. In some aspects, the time windows are defined by at least one of a sensing time duration, periodicity, or offset for transmission of measurement RS.

(172) FIG. **15** depicts aspects of an example communications device **1500**. In some aspects, communications device **1500** is a second network entity, such as BS **102** described above with respect to FIGS. **1** and **3**.

(173) The communications device **1500** includes a processing system **1505** coupled to the transceiver **1545** (e.g., a transmitter and/or a receiver) and/or a network interface **1555**. The transceiver **1545** is configured to transmit and receive signals for the communications device **1500** via the antenna **1550**, such as the various signals as described herein. The network interface **1555** is configured to obtain and send signals for the communications device **1500** via communication link(s), such as a backhaul link, midhaul link, and/or fronthaul link as described herein, such as with respect to FIG. **2**. The processing system **1505** may be configured to perform processing functions for the communications device **1500**, including processing signals received and/or to be transmitted by the communications device **1500**.

(174) The processing system **1505** includes one or more processors **1510**. In various aspects, one or more processors **1510** may be representative of one or more of receive processor **338**, transmit processor **320**, TX MIMO processor **330**, and/or controller/processor **340**, as described with respect to FIG. **3**. The one or more processors **1510** are coupled to a computer-readable medium/memory **1525** via a bus **1540**. In certain aspects, the computer-readable medium/memory **1525** is configured to store instructions (e.g., computer-executable code) that when executed by the one or more processors **1510**, cause the one or more processors **1510** to perform the method **1200** described with respect to FIG. **12**, or any aspect related to it. Note that reference to a processor of communications device **1500** performing a function may include one or more processors **1510** of communications device **1500** performing that function.

(175) In the depicted example, the computer-readable medium/memory **1525** stores code (e.g., executable instructions), such as configuration information code **1530** and configuration adjustment request code **1535**. Processing of the configuration information code **1530** and configuration adjustment request code **1535** may cause the communications device **1500** to perform the method **1200** described with respect to FIG. **12**, or any aspect related to it.

(176) The one or more processors **1510** include circuitry configured to implement (e.g., execute) the code stored in the computer-readable medium/memory **1525**, including circuitry such as configuration information circuitry **1515** and configuration adjustment request circuitry **1520**. Processing with configuration information circuitry **1515** and configuration adjustment request circuitry **1520** may cause the communications device **1500** to perform the method **1200** as described with respect to FIG. **12**, or any aspect related to it.

(177) Various components of the communications device **1500** may provide means for performing the method **1200** as described with respect to FIG. **12**, or any aspect related to it. Means for

transmitting, sending or outputting for transmission may include transceivers **332** and/or antenna(s) **334** of the BS **102** illustrated in FIG. **3** and/or the transceiver **1545** and the antenna **1550** of the communications device **1500** in FIG. **15**. Means for receiving or obtaining may include transceivers **332** and/or antenna(s) **334** of the BS **102** illustrated in FIG. **3** and/or the transceiver **1545** and the antenna **1550** of the communications device **1500** in FIG. **15**.

(178) According to some aspects, configuration information circuitry **1515** obtains information regarding a first configuration for transmission of measurement RS from a first network entity.

(179) According to some aspects, configuration adjustment request circuitry **1520** transmits a request, to the first network entity, for adjustment of the first configuration, based on a comparison of the first configuration with a second configuration for transmission of measurement RS by a second network entity or another network entity. In some aspects, the request is for adjustment of the first configuration to avoid collision of measurement RS transmissions from the first network entity with RS transmissions from the second network entity or the other network entity.

(180) In some aspects, the first configuration and the one or more second configurations indicate at least one of a measurement RS periodicity, measurement RS pattern, or measurement RS resources. In some examples, configuration information circuitry **1515** obtains the information regarding the first configuration from the first network entity. In some aspects, the first and second network entities comprise a first DU and a second DU. In some aspects, obtaining the first configuration involves signaling between the first DU and a CU, wherein at least the second DU belongs to the CU. In some aspects, the CU adjusts the first configuration based on the information regarding the one or more second configurations for transmission of measurement RS by one or more second network entities. In some aspects, if the first DU and second DU belong to different CUs, obtaining the first configuration also involves signaling between the different CUs to obtain the information regarding the one or more second configurations for transmission of measurement RS by one or more second network entities. In some aspects, obtaining the information regarding the first configuration comprises monitoring candidate measurement RS resources. In some aspects, the second network entity monitors candidate measurement resources within time windows allocated for CLI measurement. In some aspects, the time windows are defined by at least one of a sensing time duration, periodicity, or offset for transmission of measurement RS.

(181) FIG. **16** depicts aspects of an example communications device **1600**. In some aspects, communications device **1600** is a CU, such as BS **102** described above with respect to FIGS. **1** and **3**.

(182) The communications device **1600** includes a processing system **1605** coupled to the transceiver **1645** (e.g., a transmitter and/or a receiver) and/or a network interface **1655**. The transceiver **1645** is configured to transmit and receive signals for the communications device **1600** via the antenna **1650**, such as the various signals as described herein. The network interface **1655** is configured to obtain and send signals for the communications device **1600** via communication link(s), such as a backhaul link, midhaul link, and/or fronthaul link as described herein, such as with respect to FIG. **2**. The processing system **1605** may be configured to perform processing functions for the communications device **1600**, including processing signals received and/or to be transmitted by the communications device **1600**.

(183) The processing system **1605** includes one or more processors **1610**. In various aspects, one or more processors **1610** may be representative of one or more of receive processor **338**, transmit processor **320**, TX MIMO processor **330**, and/or controller/processor **340**, as described with respect to FIG. **3**. The one or more processors **1610** are coupled to a computer-readable medium/memory **1625** via a bus **1640**. In certain aspects, the computer-readable medium/memory **1625** is configured to store instructions (e.g., computer-executable code) that when executed by the one or more processors **1610**, cause the one or more processors **1610** to perform the method **1300** described with respect to FIG. **13**, or any aspect related to it. Note that reference to a processor of communications device **1600** performing a function may include one or more processors **1610** of

communications device **1600** performing that function.
(184) In the depicted example, the computer-readable medium/memory **1625** stores code (e.g., executable instructions), such as network configuration determination code **1630** and network configuration indication code **1635**. Processing of the network configuration determination code **1630** and network configuration indication code **1635** may cause the communications device **1600** to perform the method **1300** described with respect to FIG. **13**, or any aspect related to it.
(185) The one or more processors **1610** include circuitry configured to implement (e.g., execute) the code stored in the computer-readable medium/memory **1625**, including circuitry such as network configuration determination circuitry **1615** and network configuration indication circuitry **1620**. Processing with network configuration determination circuitry **1615** and network configuration indication circuitry **1620** may cause the communications device **1600** to perform the method **1300** as described with respect to FIG. **13**, or any aspect related to it.
(186) Various components of the communications device **1600** may provide means for performing the method **1300** as described with respect to FIG. **13**, or any aspect related to it. Means for transmitting, sending or outputting for transmission may include transceivers **332** and/or antenna(s) **334** of the BS **102** illustrated in FIG. **3** and/or the transceiver **1645** and the antenna **1650** of the communications device **1600** in FIG. **16**. Means for receiving or obtaining may include transceivers **332** and/or antenna(s) **334** of the BS **102** illustrated in FIG. **3** and/or the transceiver **1645** and the antenna **1650** of the communications device **1600** in FIG. **16**.
(187) According to some aspects, network configuration determination circuitry **1615** determines a first configuration for transmission of measurement RS from a first network entity, based on information regarding one or more second configurations for transmission of measurement RS from one or more second network entities.
(188) According to some aspects, network configuration indication circuitry **1620** transmits information regarding the first configuration to the first network entity. In some examples, network configuration indication circuitry **1620** transmits information regarding the second configuration to the second network entity.
(189) In some aspects, the first configuration and the one or more second configurations indicate at least one of a measurement RS periodicity, measurement RS pattern, or measurement RS resources. In some aspects, determining the first configuration comprises adjusting the first configuration to avoid collision of measurement RS transmissions from the first network entity with RS transmissions from the one or more second network entities. In some aspects, determining the first configuration comprises adjusting the first configuration to enable measurement RS transmissions from the first network entity to be received by multiple of the one or more second network entities. In some aspects, the first and second network entities comprise first and second DU.
EXAMPLE CLAUSES
(190) Implementation examples are described in the following numbered clauses:
(191) Clause 1: A method for wireless communications by a first network entity, comprising: obtaining a first configuration for transmission of measurement RS by the first network entity, wherein the first configuration is adjusted based on information regarding one or more second configurations for transmission of measurement RS by one or more second network entities; and participating in a CLI measurement procedure, in accordance with the first configuration.
(192) Clause 2: The method of Clause 1, wherein participating in the CLI measurement procedure comprises at least one of: scheduling measurement RS transmissions, in accordance with the first configuration; or scheduling measurement RS reception, in accordance with the first configuration.
(193) Clause 3: The method of any one of Clauses 1 and 2, wherein the first configuration and the one or more second configurations indicate at least one of a measurement RS periodicity, measurement RS pattern, or measurement RS resources.
(194) Clause 4: The method of any one of Clauses 1-3, further comprising: adjusting the first configuration, based on the information regarding the one or more second configurations, to avoid

collision of measurement RS transmissions from the first network entity with RS transmissions from the one or more second network entities.
(195) Clause 5: The method of any one of Clauses 1-4, further comprising: adjusting the first configuration, based on the information regarding the one or more second configurations, to enable measurement RS transmissions from the first network entity to be received by multiple of the one or more second network entities.
(196) Clause 6: The method of any one of Clauses 1-5, further comprising: obtaining the information regarding the one or more second configurations from the one or more second network entities.
(197) Clause 7: The method of Clause 6, further comprising: transmitting information regarding the first configuration to one or more of the second network entities; receiving a request, from one or more of the second network entities, to adjust the first configuration; and adjusting the first configuration in response to the request.
(198) Clause 8: The method of any one of Clauses 1-7, wherein the first and second network entities comprise a first DU and a second DU.
(199) Clause 9: The method of Clause 8, wherein: obtaining the first configuration involves signaling between the first DU and a CU via a backhaul interface, wherein the first DU belongs to the CU; and the CU adjusts the first configuration based on the information regarding the one or more second configurations for transmission of measurement RS by one or more second network entities.
(200) Clause 10: The method of Clause 9, wherein if the first DU and the second DU belong to different CUs, obtaining the first configuration also involves signaling between the different CUs to obtain the information regarding the one or more second configurations for transmission of measurement RS by one or more second network entities.
(201) Clause 11: The method of any one of Clauses 1-10, further comprising: obtaining the information regarding the one or more second configurations by monitoring candidate measurement RS resources.
(202) Clause 12: The method of Clause 11, wherein participating in the CLI measurement procedure comprises: transmitting measurement RS on a candidate measurement RS resources after determining that candidate measurement RS resource is unoccupied by any of the one or more second network entities.
(203) Clause 13: The method of Clause 11, wherein the first network entity monitors candidate measurement resources within time windows allocated for CLI measurement.
(204) Clause 14: The method of Clause 13, wherein the time windows are defined by at least one of a sensing time duration, periodicity, or offset for transmission of measurement RS.
(205) Clause 15: A method for wireless communications by a second network entity, comprising: obtaining information regarding a first configuration for transmission of measurement RS from a first network entity; and transmitting a request, to the first network entity, for adjustment of the first configuration, based on a comparison of the first configuration with a second configuration for transmission of measurement RS by a second network entity or another network entity.
(206) Clause 16: The method of Clause 15, wherein the first configuration and the one or more second configurations indicate at least one of a measurement RS periodicity, measurement RS pattern, or measurement RS resources.
(207) Clause 17: The method of any one of Clauses 15 and 16, wherein the request is for adjustment of the first configuration to avoid collision of measurement RS transmissions from the first network entity with RS transmissions from the second network entity or the other network entity.
(208) Clause 18: The method of any one of Clauses 15-17, further comprising: obtaining the information regarding the first configuration from the first network entity.
(209) Clause 19: The method of any one of Clauses 15-18, wherein the first and second network

entities comprise a first DU and a second DU.
(210) Clause 20: The method of Clause 19, wherein: obtaining the first configuration involves signaling between the first DU and a CU, wherein at least the second DU belongs to the CU; and the CU adjusts the first configuration based on the information regarding the one or more second configurations for transmission of measurement RS by one or more second network entities.
(211) Clause 21: The method of Clause 20, wherein if the first DU and second DU belong to different CUs, obtaining the first configuration also involves signaling between the different CUs to obtain the information regarding the one or more second configurations for transmission of measurement RS by one or more second network entities.
(212) Clause 22: The method of any one of Clauses 15-21, wherein obtaining the information regarding the first configuration comprises monitoring candidate measurement RS resources.
(213) Clause 23: The method of Clause 22, wherein the second network entity monitors candidate measurement resources within time windows allocated for CLI measurement.
(214) Clause 24: The method of Clause 23, wherein the time windows are defined by at least one of a sensing time duration, periodicity, or offset for transmission of measurement RS.
(215) Clause 25: A method for wireless communications by a CU, comprising: determining a first configuration for transmission of measurement RS from a first network entity, based on information regarding one or more second configurations for transmission of measurement RS from one or more second network entities; and transmitting information regarding the first configuration to the first network entity.
(216) Clause 26: The method of Clause 25, further comprising: transmitting information regarding the second configuration to the second network entity.
(217) Clause 27: The method of any one of Clauses 25 and 26, wherein the first configuration and the one or more second configurations indicate at least one of a measurement RS periodicity, measurement RS pattern, or measurement RS resources.
(218) Clause 28: The method of any one of Clauses 25-27, wherein determining the first configuration comprises adjusting the first configuration to avoid collision of measurement RS transmissions from the first network entity with RS transmissions from the one or more second network entities.
(219) Clause 29: The method of any one of Clauses 25-28, wherein determining the first configuration comprises adjusting the first configuration to enable measurement RS transmissions from the first network entity to be received by multiple of the one or more second network entities.
(220) Clause 30: The method of any one of Clauses 25-29, wherein the first and second network entities comprise first and second DU.
(221) Clause 31: A processing system, comprising: a memory comprising computer-executable instructions; one or more processors configured to execute the computer-executable instructions and cause the processing system to perform a method in accordance with any one of Clauses 1-30.
(222) Clause 32: A processing system, comprising means for performing a method in accordance with any one of Clauses 1-30.
(223) Clause 33: A non-transitory computer-readable medium comprising computer-executable instructions that, when executed by one or more processors of a processing system, cause the processing system to perform a method in accordance with any one of Clauses 1-30.
(224) Clause 34: A computer program product embodied on a computer-readable storage medium comprising code for performing a method in accordance with any one of Clauses 1-30.

ADDITIONAL CONSIDERATIONS

(225) The preceding description is provided to enable any person skilled in the art to practice the various aspects described herein. The examples discussed herein are not limiting of the scope, applicability, or aspects set forth in the claims. Various modifications to these aspects will be readily apparent to those skilled in the art, and the general principles defined herein may be applied to other aspects. For example, changes may be made in the function and arrangement of elements

discussed without departing from the scope of the disclosure. Various examples may omit, substitute, or add various procedures or components as appropriate. For instance, the methods described may be performed in an order different from that described, and various actions may be added, omitted, or combined. Also, features described with respect to some examples may be combined in some other examples. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method that is practiced using other structure, functionality, or structure and functionality in addition to, or other than, the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.

(226) The various illustrative logical blocks, modules and circuits described in connection with the present disclosure may be implemented or performed with a general purpose processor, a digital signal processor (DSP), an ASIC, a field programmable gate array (FPGA) or other programmable logic device (PLD), discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any commercially available processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, a system on a chip (SoC), or any other such configuration.

(227) As used herein, a phrase referring to "at least one of" a list of items refers to any combination of those items, including single members. As an example, "at least one of: a, b, or c" is intended to cover a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiples of the same element (e.g., a-a, a-a-a, a-a-b, a-a-c, a-b-b, a-c-c, b-b, b-b-b, b-b-c, c-c, and c-c-c or any other ordering of a, b, and c).

(228) As used herein, the term "determining" encompasses a wide variety of actions. For example, "determining" may include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Also, "determining" may include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, "determining" may include resolving, selecting, choosing, establishing and the like.

(229) The methods disclosed herein comprise one or more actions for achieving the methods. The method actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of actions is specified, the order and/or use of specific actions may be modified without departing from the scope of the claims. Further, the various operations of methods described above may be performed by any suitable means capable of performing the corresponding functions. The means may include various hardware and/or software component(s) and/or module(s), including, but not limited to a circuit, an application specific integrated circuit (ASIC), or processor.

(230) The following claims are not intended to be limited to the aspects shown herein, but are to be accorded the full scope consistent with the language of the claims. Within a claim, reference to an element in the singular is not intended to mean "one and only one" unless specifically so stated, but rather "one or more." Unless specifically stated otherwise, the term "some" refers to one or more. No claim element is to be construed under the provisions of 35 U.S.C. § 112(f) unless the element is expressly recited using the phrase "means for". All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims.

Claims

1. A method of wireless communication by a first network entity, comprising: obtaining a first configuration for transmission of measurement reference signals (RS) by the first network entity, wherein the first configuration is adjusted based on information regarding one or more second configurations for transmission of measurement RS by one or more second network entities; and participating in a cross link interference (CLI) measurement procedure, in accordance with the first configuration.
2. The method of claim 1, wherein participating in the CLI measurement procedure comprises at least one of: scheduling measurement RS transmissions, in accordance with the first configuration; or scheduling measurement RS reception, in accordance with the first configuration.
3. The method of claim 1, wherein the first configuration and the one or more second configurations indicate at least one of a measurement RS periodicity, measurement RS pattern, or measurement RS resources.
4. The method of claim 1, further comprising adjusting the first configuration, based on the information regarding the one or more second configurations, to avoid collision of measurement RS transmissions from the first network entity with RS transmissions from the one or more second network entities.
5. The method of claim 1, further comprising adjusting the first configuration, based on the information regarding the one or more second configurations, to enable measurement RS transmissions from the first network entity to be received by multiple of the one or more second network entities.
6. The method of claim 1, further comprising obtaining the information regarding the one or more second configurations from the one or more second network entities.
7. The method of claim 6, further comprising: transmitting information regarding the first configuration to one or more of the second network entities; receiving a request, from one or more of the second network entities, to adjust the first configuration; and adjusting the first configuration in response to the request.
8. The method of claim 1, wherein: the first network entity and the second network entities comprise a first distributed unit (DU) and a second DU.
9. The method of claim 8, wherein: obtaining the first configuration involves signaling between the first DU and a central unit (CU) via a backhaul interface, wherein the first DU belongs to the CU; and the CU adjusts the first configuration based on the information regarding the one or more second configurations for transmission of measurement RS by one or more second network entities.
10. The method of claim 9, wherein: if the first DU and second DU belong to different CUs, obtaining the first configuration also involves signaling between the different CUs to obtain the information regarding the one or more second configurations for transmission of measurement RS by one or more second network entities.
11. The method of claim 1, further comprising obtaining the information regarding the one or more second configurations by monitoring candidate measurement RS resources.
12. The method of claim 11, wherein participating in the CLI measurement procedure comprises: transmitting measurement RS on a candidate measurement RS resources after determining that candidate measurement RS resource is unoccupied by any of the one or more second network entities.
13. The method of claim 11, wherein the first network entity monitors candidate measurement resources within time windows allocated for CLI measurement.
14. The method of claim 13, wherein the time windows are defined by at least one of a sensing time duration, periodicity, or offset for transmission of measurement RS.

15. A method of wireless communication by a second network entity, comprising: obtaining information regarding a first configuration for transmission of measurement reference signals (RS) by a first network entity; and transmitting a request, to the first network entity, for adjustment of the first configuration, based on a comparison of the first configuration with a second configuration for transmission of measurement RS by the second network entity or another network entity.
16. The method of claim 15, wherein the first configuration and the second configuration indicate at least one of a measurement RS periodicity, measurement RS pattern, or measurement RS resources.
17. The method of claim 15, wherein the request is for adjustment of the first configuration to avoid collision of measurement RS transmissions from the first network entity with RS transmissions from the second network entity or the other network entity.
18. The method of claim 15, further comprising obtaining, from the first network entity, the information regarding the first configuration.
19. The method of claim 15, wherein: the first network entity and the second network entity comprise a first distributed unit and a second distributed unit (DU).
20. The method of claim 19, wherein: obtaining the first configuration involves signaling between the first DU and a central unit (CU), wherein at least the second DU belongs to the CU; and the CU adjusts the first configuration based on the information regarding the second configuration for transmission of measurement RS by one or more second network entities.
21. The method of claim 20, wherein: if the first DU and second DU belong to different CUs, obtaining the first configuration also involves signaling between the different CUs to obtain the information regarding the second configuration for transmission of measurement RS by one or more second network entities.
22. The method of claim 15, wherein: obtaining the information regarding the first configuration comprises monitoring candidate measurement RS resources.
23. The method of claim 22, wherein the second network entity monitors candidate measurement resources within time windows allocated for CLI measurement.
24. The method of claim 23, wherein the time windows are defined by at least one of a sensing time duration, periodicity, or offset for transmission of measurement RS.
25. A method of wireless communication by a central unit (CU), comprising: determining a first configuration for transmission of measurement reference signals (RS) by a first network entity, based on information regarding one or more second configurations for transmission of measurement RS by one or more second network entities, wherein determining the first configuration comprises adjusting the first configuration to avoid collision of measurement RS transmissions from the first network entity with RS transmissions from the one or more second network entities; and transmitting information regarding the first configuration to the first network entity.
26. The method of claim 25, further comprising transmitting information regarding the one or more second configurations to the second network entities.
27. The method of claim 25, wherein the first configuration and the one or more second configurations indicate at least one of a measurement RS periodicity, measurement RS pattern, or measurement RS resources.
28. The method of claim 25, wherein determining the first configuration comprises adjusting the first configuration to enable measurement RS transmissions from the first network entity to be received by multiple of the one or more second network entities.
29. The method of claim 25, wherein: the first network entity and the second network entities comprise first and second distributed units (DUs).